THE ROLE OF SLEEP IN THE RECRUITMENT OF NEWLY LEARNED FISH AND FLOWER WORDS INTO UNCONSCIOUS PROCESSING SYSTEMS

Julia Trudeau, Elizabeth Hecht, Eylul Akgul, Paola Cervantes, Samantha Cheng, Richard L. Abrams, Ph.D.

Psychology Department, Loyola Marymount University

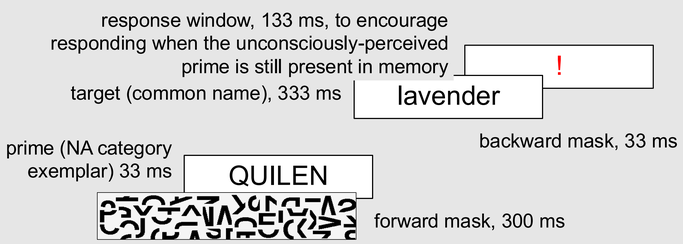
Introduction

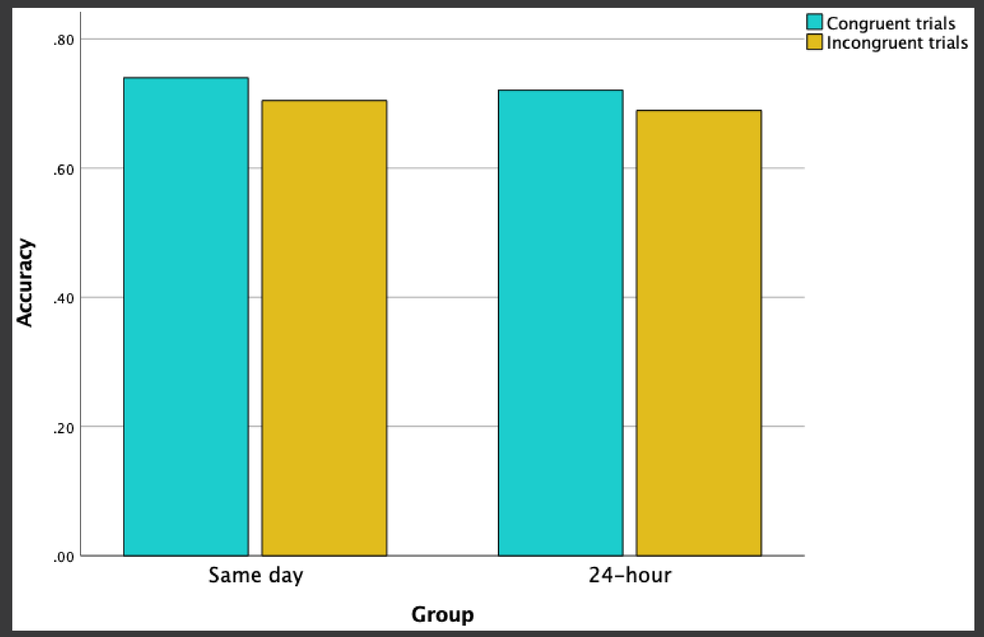
- Unconscious semantic priming is a phenomenon in which a briefly flashed, consciously imperceptible word influences the categorization of a different target word that participants are instructed to attend to.
- Newly acquired words can be integrated into unconscious processing systems as soon as 24 hours from the learning period. The priming effect can last up until 48 hours after the learning period (Morgan & Abrams, 2021).
- The Morgan & Abrams experiment hypothesized that sleep may play a key role in the integration of novel words into unconscious processing systems.
- Our study focused on the recruitment of newly-acquired words under display conditions typically associated with unconscious processing.
- Over the course of our experiment, the acquisition of rare and presumed-to-be unfamiliar exemplars of two common categories, fish and flower, was examined. We manipulated the duration between practice and priming task sessions, either done on the same day or 24 hours after.
- Our objective was to evaluate the process by which newly-acquired vocabulary is recruited into unconscious processing and to examine the role of sleep in the consolidation of these newly-acquired words.

Method

- All experiments began with a session of extensive learning (30-45 minutes) in which participants were introduced to the novel exemplars first by pairing the words with pictures of the corresponding fish or flower, then practiced categorizing the words alone.
- Participants were split into two groups. The same day group performed the priming session immediately after the practice session. The 24-hour group performed the priming session 24 hours after the practice session.
- During this priming task, the newly-acquired (NA) exemplars acted as the masked primes, and common fish and flower names acted as the visible targets.

Trial sequence in priming tasks
(Instruction was to categorize the target)





Results

- $N = 41$
- Same day group
  - $N = 19$, $t\ (18) = 2.54$, $p = 0.021$
  - Congruent $M = 0.74$
  - Incongruent $M = 0.70$
- 24-hour group
  - $N = 22$, $t\ (21) = 2.05$, $p = 0.053$
  - Congruent $M = 0.72$
  - Incongruent $M = 0.69$
- The same day group showed significant priming
- The 24-hour group showed near significant priming

Discussion

- These experiments demonstrate that novel category exemplars are rapidly recruited into systems that handle unconscious processing of visual language.
- Similar to overlearned words, newly acquired words function as unconscious primes within an hour of learning.
- Priming effects of newly acquired words can last for at least 24 hours after the initial learning session.
- Sleep has no effect on the integration of newly acquired words into unconscious processing systems
- Further research would examine how long this priming effect can last and if it can remain indefinitely in memory. For example, what would happen if participants come back a week later?
- Future research should also answer the question: What is the magnitude of function as a prime?

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